SAT Math Mastery · Session 1

Quadratics and Nonlinear Functions

Time	3 hours: Part A in class (140 min solving) + 40 min self-check and error log. Part B is the self-assessment, done independently.
Topics	parabolas: vertex, zeros and intercepts · function notation and composition · transformations · exponential and power functions · direct, inverse and joint variation · equivalent forms · undefined values
Name / Date	_____ _____

PART A · IN CLASS — 35 PROBLEMS

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal). Grey boxes show exactly what to type in Desmos, one expression per line; tap a grey dot to read an intersection, intercept or vertex. A letter such as a, c or k becomes a slider when you accept Desmos's offer to add one.

1. HARD · DESMOS

The function g satisfies $g(kn) = n + 10$ for every real number n, where k is a constant. If $g(4) = 82$, what is the value of k?

1. Set $kn = 4$, so $n = \frac{4}{k}$, and the rule gives $\frac{4}{k} + 10 = 82$.

2. Type `y=4/x+10` and `y=82`.

3. Tap the intersection: (0.0556, 82). Its x-value is k.

4. Type `4/72`: it returns 0.0556, which is $\frac{1}{18}$.

Answer: $\frac{1}{18}$

2. HARD · DESMOS

A decorative fountain in a public park shoots a jet of water whose height above the ground, in meters, is modeled by

$$h(t) = -4\,(t-3)^2 + 45,$$

where t is the time, in seconds, after the jet is activated. For which values of t does the water reach a height of 29 meters?

(A) $t = 5$ only

(B) $t = 1$ and $t = 5$ ✓

(C) $t = 2$ and $t = 4$

(D) $t = 1$ only

1. Type `y=-4(x-3)^2+45` and the line `y=29`.

2. Tap the two grey dots: (1, 29) and (5, 29).

3. The height is 29 at $t = 1$ and at $t = 5$, so both values count.

Answer: (B)

3. HARD · DESMOS

The function f is defined by

$$f(x) = a\,(x-2)\,(x+6)\,,$$

where a is a constant. The graph of $y = f(x)$ in the xy-plane passes through the point $(0, -24)$. What is the value of $f(5)$?

1. Type `y=a(x-2)(x+6)` (accept the slider for a) and plot the point `(0,-24)`.

2. Slide a until the curve passes through the point: $a = 2$.

3. Type `2(5-2)(5+6)`: it returns 66.

Answer: 66

4. HARD · DESMOS

The function f is defined by $f(x) = a \cdot 2^x$, where a is a constant. If $f(3) = 40$, what is the value of $f(5)$?

(A) 64

(B) 128

(C) 160 ✓

(D) 200

1. Type `y=a*2^x` (slider a) and plot the point `(3,40)`.

2. Slide a until the curve passes through the point: $a = 5$.

3. Type `5*2^5`: it returns 160.

Answer: (C)

5. HARD · DESMOS

A ball is thrown upward, and its height above the ground, in meters, is modeled by

$$h(t) = -5\,(t-2)^2 + 30,$$

where t is the time, in seconds, after the ball is thrown. For which values of t does the ball reach a height of 10 meters?

(A) $t = 0$ and $t = 4$ ✓

(B) $t = 1$ and $t = 3$

(C) $t = 0$ only

(D) $t = 4$ only

1. Type `y=-5(x-2)^2+30` and the line `y=10`.

2. Tap the two grey dots: (0, 10) and (4, 10).

3. The height is 10 at $t = 0$ and at $t = 4$.

Answer: (A)

6. HARD · DESMOS

The function h is defined by $h(x) = (x - a)(x - b)$, where a and b are constants and $a < b$. The graph of $y = h(x)$ crosses the x-axis at $x = -3$ and $x = 7$. What is the minimum value of h?

(A) -20 (B) -10

(C) 25 **(D) -25 ✓**

1. The zeros are -3 and 7: type `y=(x+3)(x-7)`.

2. Tap the vertex (the lowest point): $(2, -25)$.

3. The minimum value of h is -25.

Answer: (D)

7. HARD · DESMOS

The function p is defined by $p(x) = a \cdot x^3$, where a is a nonzero constant. If $p(2) = 40$, what is the value of $p(-2)$?

1. Type `40/2^3`: it returns 5, so $a = 5$.

2. Type `p(x)=5x^3` and then `p(-2)`: it returns -40.

3. An odd power flips the sign, so $p(-2) = -p(2) = -40$.

Answer: -40

8. HARD · DESMOS

The function N is defined by $N(t) = k \cdot 2^{-at}$ for positive constants k and a. The table shows values of N for select values of t.

t	0	1	2
$N(t)$	200	50	12.5

What is the value of a?

1. From $N(0) = 200$ we get $k = 200$. Type `y=200*2^(-a*x)` (slider a).

2. Plot the points `(1,50)` and `(2,12.5)`.

3. Slide a until the curve passes through both points: $a = 2$.

Answer: 2

9. HARD · DESMOS

The variable z varies jointly with x and the square of y. If $z = 48$ when $x = 3$ and $y = 4$, what is the value of z when $x = 5$ and $y = 2$?

(A) 10

(B) 16

(C) 20 ✓

(D) 25

1. Write $z = kxy^2$. Type `48/(3*4^2)` : it returns $k = 1$.

2. Type `1*5*2^2` : it returns 20.

3. So $z = 20$ when $x = 5$ and $y = 2$.

Answer: (C)

10. HARD · DESMOS

The graph of $y = h(x)$ is the graph of $y = x^2$ shifted right 3 units and down 5 units. What is the value of $h(7)$?

(A) 6

(B) 11 ✓

(C) 14

(D) 16

1. Right 3 and down 5 gives $h(x) = (x - 3)^2 - 5$. Type `h(x)=(x-3)^2-5` .

2. Type `h(7)` : it returns 11.

3. So $h(7) = 11$.

Answer: (B)

11. HARD · DESMOS

A ball's height above the ground, in meters, t seconds after being launched from a platform is modeled by $h(t) = -5t^2 + 20t + c$, where c is the platform height. If $h(1) = 35$, what is the value of $h(3)$?

1. Type `y=-5x^2+20x+c` (slider c) and plot the point `(1,35)` .

2. Slide c until the curve passes through the point: $c = 20$.

3. Type `-5*3^2+20*3+20` : it returns 35.

Answer: 35

12. HARD · DESMOS

The function g is defined by

$$g(x) = a\,(x-3)\,(x+4)\,,$$

where a is a constant. The graph of $y = g(x)$ in the xy-plane passes through the point $(0, -36)$. What is the value of $g(6)$?

1. Type `y=a(x-3)(x+4)` (slider a) and plot the point `(0,-36)`.

2. Slide a until the curve passes through the point: $a = 3$.

3. Type `3(6-3)(6+4)` : it returns 90.

Answer: 90

13. HARD · DESMOS

A cooperative tracks its annual export tonnage with the function $f(x) = a \cdot 3^x$, where a is a constant and x is the number of years since record-keeping began. If $f(2) = 54$, what is the value of $f(4)$?

(A) 108

(B) 162

(C) 324

(D) 486 ✓

1. Type `y=a*3^x` (slider a) and plot the point `(2,54)`.

2. Slide a until the curve passes through the point: $a = 6$.

3. Type `6*3^4` : it returns 486.

Answer: (D)

14. HARD · DESMOS

The height, in meters, of a ball above the ground t seconds after being thrown is given by $h(t) = -4.9t^2 + 14.7t + 2$. Which of the following is closest to the number of seconds it takes for the ball to hit the ground ($h(t) = 0$)?

(A) 3.1 ✓

(B) 3.6

(C) 2.1

(D) 2.6

1. Type `y=-4.9x^2+14.7x+2`.

2. Tap the positive x-intercept: (3.13, 0).

3. Of the options, 3.1 is the closest.

Answer: (A)

15. HARD · DESMOS

The graph of the function $g(x) = (x-3)^2 - 7$ is the same as the graph of $y = x^2$ after which of the following transformations?

(A) Shifted left 3 and up 7

(B) Shifted right 3 and down 7 ✓

(C) Shifted right 3 and up 7

(D) Shifted left 3 and down 7

1. Type `y=(x-3)^2-7` and `y=x^2`.

2. Tap the vertex of each: $(3, -7)$ and $(0, 0)$.

3. The vertex moved 3 to the right and 7 down.

Answer: (B)

16. HARD · DESMOS

If $x = 5$ is one solution to the equation $x^2 - kx + 20 = 0$, what is the value of k?

1. Put $x = 5$ into the equation: $25 - 5k + 20 = 0$. Let x stand for k: type `y=25-5x+20` and `y=0`.

2. Tap the intersection: $(9, 0)$.

3. So $k = 9$.

Answer: 9

17. HARD · DESMOS

A company's weekly revenue, in thousands of dollars, from selling a product priced at x dollars is modeled by $R(x) = -3x^2 + 150x$. What price x maximizes the weekly revenue?

(A) 15

(B) 20

(C) 25 ✓

(D) 30

1. Type `y=-3x^2+150x`.

2. Tap the vertex: $(25, 3750)$.

3. The revenue is greatest at the price $x = 25$.

Answer: (C)

18. HARD · DESMOS

The equation $x^2 - 12x + c = 0$ has two equal (repeated) roots. What is the value of c?

1. Type `y=x^2-12x+c` (slider c).

2. Slide c until the parabola just touches the x-axis at one point.

3. That happens at $c = 36$, where the vertex is $(6, 0)$.

Answer: 36

19. HARD · DESMOS

The function f is defined by $f(x) = \dfrac{12 - x}{2 + x}$, where $x \neq -2$. As the value of x increases without bound, the value of $f(x)$ approaches which of the following?

(A) -1 ✓ (B) 1

(C) 12 (D) -2

1. Type `y=(12-x)/(2+x)` and zoom out to the right.

2. The curve flattens along a horizontal line. Type `y=-1` to confirm it hugs that line.

3. As x increases without bound, $f(x)$ approaches -1.

Answer: (A)

20. HARD · DESMOS

$y = (x - p)^2 (x + p)(x - q)$, where p and q are positive constants and $p \neq q$. How many distinct x-intercepts does the graph of this equation have in the xy-plane?

(A) 4 (B) 1

(C) 2 **(D)** 3 ✓

1. Choose $p = 1$ and $q = 2$: type `y=(x-1)^2(x+1)(x-2)`.

2. Tap the x-intercepts: -1, 1 and 2.

3. The graph has 3 distinct x-intercepts (the repeated factor adds no new intercept).

Answer: (D)

21. HARD · DESMOS

The table below shows some values of the function N.

t	0	1	2
$N(t)$	100	25	$\frac{25}{4}$

If $N(t) = k \cdot 2^{-at}$ for positive constants k and a, what is the value of a?

1. From $N(0) = 100$ we get $k = 100$. Type `y=100*2^(-a*x)` (slider a).

2. Plot the points `(1,25)` and `(2,6.25)`.

3. Slide a until the curve passes through both points: $a = 2$.

Answer: 2

22. HARD · DESMOS

The function f is defined by $f(x) = 3x + k$, where k is a constant. If $f(f(2)) = 38$, what is the value of k?

1. Type `f(x)=3x+k`: $f(2) = 6 + k$, and $f(f(2)) = 3(6 + k) + k = 18 + 4k$.

2. Type `y=18+4x` and `y=38` (letting x stand for k).

3. Tap the intersection: (5, 38), so $k = 5$.

Answer: 5

23. HARD · DESMOS

The function f is defined by $f(x) = a(x+4)(x-10)$, where a is a constant. The minimum value of f is -98. What is the value of $f(0)$?

(A) -40

(B) 80

(C) -98 ✓

(D) -80

1. Type `y=a(x+4)(x-10)` (slider a).

2. Slide a until the lowest point has $y = -98$: $a = 2$.

3. Type `2*4*(-10)`: it returns $f(0) = -80$.

Answer: (C)

24. HARD · DESMOS

A theater's revenue, in dollars, from selling t tickets is $R(t) = t(320 - 10t)$. For which of the following values of t is the revenue 2,400 dollars?

(A) 26

(B) 14 ✓

(C) 18

(D) 20

1. Type `y=x(320-10x)` and the line `y=2400`.
2. Tap the two grey dots: (12, 2400) and (20, 2400).
3. Only 20 appears among the options.

Answer: (B)

25. HARD · DESMOS

$y = \dfrac{5(x+3)(x-9)(x+2)}{(x-5)(x-5c)}$ In the equation above, c is a positive constant. The expression is undefined at $x = 10$ (and $x = 5$). What is the value of c?

(A) -2

(B) 2

(C) 3

(D) -3 ✓

1. The expression is undefined when $x - 5c = 0$. Type `y=x-5c` (slider c).
2. Slide c until the line crosses the x-axis at $x = 10$: $c = 2$.
3. Type `10-5*2`: it returns 0, confirming $c = 2$.

Answer: (D)

26. HARD · DESMOS

The function $f(x) = ax^2 + bx + c$ has $a > 0$ and $c < 0$. Which of the following must be true?
I. f has two distinct real zeros
II. The zeros of f have opposite signs
III. The vertex of the graph of f lies in the fourth quadrant

(A) I only ✓

(B) I and III only

(C) I and II only

(D) I, II and III

1. Try an example with $a > 0$ and $c < 0$: type `y=x^2+x-6`.
2. Tap the intercepts $(-3, 0)$ and $(2, 0)$ and the vertex $(-0.5, -6.25)$.
3. Statements I and II hold (and the product of zeros is $\dfrac{c}{a} < 0$ for every such f); here the vertex is in the third quadrant, so III does not have to hold.

Answer: (A)

27. HARD · DESMOS

x	$g(x)$
-12	$-\frac{17}{4}$
-9	$-\frac{10}{3}$
3	258

The table above shows three values of x and their corresponding values of $g(x)$. If $g(x) = \frac{x^2 + a}{bx} + 2^{(cx-1)}$, what is the value of abc?

1. At $x = 3$: $258 = 2 + 2^8$, so $c = 3$. Type `2^(3*3-1)` : it returns 256, leaving $\frac{9 + a}{3b} = 2$, i.e. $9 + a = 6b$.

2. At $x = -12$ the term 2^{-37} is almost 0. Type `2^(-37)` to see it, so $\frac{144 + a}{-12b} = -\frac{17}{4}$, i.e. $144 + a = 51b$.

3. Type `y=6x-9` and `y=51x-144` (letting x stand for b). Tap the intersection: (3, 9), so $b = 3$ and $a = 9$.

4. Check $x = -9$: type `(81+9)/(-27)` ; it returns -3.333, which is $-\frac{10}{3}$. Then type `9*3*3` : it returns 81.

Answer: 81

28. HARD · DESMOS

The function f is defined by $f(x) = 5(3)^{x+2}$. The functions g and h are equivalent to f. Which of the following equations displays, as a constant or coefficient, the y-coordinate of the y-intercept of the graph of the function it defines in the xy-plane?

I. $g(x) = 45(3)^x$

II. $h(x) = 15(3)^{x+1}$

(A) Neither I nor II

(B) I only ✓

(C) II only

(D) I and II

1. Type `y=5*3^(x+2)` and tap the y-intercept: (0, 45).

2. Type `y=45*3^x` : it lies exactly on the graph, and the coefficient 45 is the intercept.

3. Type `y=15*3^(x+1)` : it also lies on the graph, but its coefficient is 15, not 45.

4. Only the first form displays the intercept.

Answer: (B)

29. HARD · DESMOS

The function f is defined by $f(x) = 96(2)^{x-3}$. The functions g and h are equivalent to f. Which of the following equations displays, as a constant or coefficient, the y-coordinate of the y-intercept of the graph of the function it defines in the xy-plane?
I. $g(x) = 48(2)^{x-2}$
II. $h(x) = 12(2)^{x}$

(A) I and II
(B) Neither I nor II
(C) I only ✓
(D) II only

1. Type `y=96*2^(x-3)` and tap the y-intercept: (0, 12).
2. Type `y=48*2^(x-2)` : same graph, but its coefficient is 48, not 12.
3. Type `y=12*2^x` : same graph, and the coefficient 12 is the intercept.
4. Only the second form displays the intercept.
Answer: (C)

30. HARD · DESMOS

The function f is defined by $f(x) = 3(4)^{x+2}$, where $x \geq 0$. The functions g and h are equivalent to f. Which of the following equations displays, as a constant or coefficient, the minimum value of the function it defines, where $x \geq 0$?
I. $g(x) = 48(4)^{x}$
II. $h(x) = 48(2)^{2x}$

(A) I only ✓
(B) II only
(C) I and II
(D) Neither I nor II

1. Type `y=3*4^(x+2){x>=0}` and tap the lowest point, at $x = 0$: (0, 48).
2. Type `y=48*4^x` : same graph, and the coefficient 48 is the minimum.
3. Type `y=48*2^(2x)` : same graph again, with coefficient 48.
4. Both forms display the minimum.
Answer: (A)

31. Hard · Desmos

The function f is defined by $f(x) = 6(2)^{x+3}$, where $x \geq 0$. The functions g and h are equivalent to f. Which of the following equations displays, as a constant or coefficient, the minimum value of the function it defines, where $x \geq 0$?
I. $g(x) = 12(2)^{x+2}$
II. $h(x) = 24(2)^{x+1}$

(A) II only

(B) I and II

(C) Neither I nor II

(D) I only ✓

1. Type `y=6*2^(x+3){x>=0}` and tap the lowest point, at $x = 0$: (0, 48).

2. Type `y=12*2^(x+2)` : same graph, but its coefficient is 12, not 48.

3. Type `y=24*2^(x+1)` : same graph, but its coefficient is 24, not 48.

4. Neither form displays the minimum.

Answer: (D)

32. Hard · Desmos

The function $f(x) = 200(1.5)^{\frac{x}{2}}$ is defined for all x. For every increase of x by 4, the value of $f(x)$ increases by p%. What is the value of p?

1. Type `f(x)=200*1.5^(x/2)` .

2. Type `f(4)/f(0)` : it returns 2.25, the growth factor for every increase of 4.

3. Type `(2.25-1)*100` : it returns 125.

Answer: 125

33. Hard · Desmos

The function $f(t) = 2400(0.8)^{\frac{t}{5}}$ models the value, in dollars, of a machine t years after it was bought. For every increase of t by 10, the value of $f(t)$ decreases by p%. What is the value of p?

(A) 40

(B) 64

(C) 20 ✓

(D) 36

1. Type `f(t)=2400*0.8^(t/5)` .

2. Type `f(10)/f(0)` : it returns 0.64, the factor for every increase of 10.

3. Type `(1-0.64)*100` : it returns 36, so the value decreases by 36%.

Answer: (C)

34. HARD · DESMOS

The function $N(t) = 500(2)^{\frac{t}{3}}$ models the number of bacteria in a culture t hours after the start. Which of the following gives the number of bacteria m minutes after the start?

(A) $N(m) = 500(2)^{\frac{m}{20}}$

(B) $N(m) = 500(2)^{\frac{m}{180}}$ ✓

(C) $N(m) = 500(2)^{180m}$

(D) $N(m) = 500(2)^{20m}$

1. m minutes is $\dfrac{m}{60}$ hours, so $N = 500 \cdot 2^{\frac{m}{60}/3}$. Type `N(t)=500*2^(t/3)`.

2. Test $m = 180$ minutes (3 hours): type `N(3)`; it returns 1000.

3. Type `500*2^(180/180)`: it returns 1000, a match. Type `500*2^(180/20)`: it returns 256000, so the $\dfrac{m}{20}$ form fails.

4. The exponent must be $\dfrac{m}{180}$.

Answer: (B)

35. HARD · DESMOS

The function $f(x) = 12{,}000(0.85)^x$ models the number of visitors a museum had each year, where x is the number of years since the start of 2018 and $0 \le x \le 6$. If $y = f(x)$ is graphed in the xy-plane, which of the following is the best interpretation of the y-intercept of the graph in this context?

(A) The minimum estimated number of visitors during the 6 years was 12,000.

(B) The estimated number of visitors in 2018 was 4,526.

(C) The estimated number of visitors in 2018 was 12,000.

(D) The minimum estimated number of visitors during the 6 years was 4,526. ✓

1. Type `y=12000*0.85^x{0<=x<=6}`.

2. Tap the y-intercept: (0, 12000). The input $x = 0$ is the start of 2018.

3. So 12,000 is the estimated number of visitors in 2018.

Answer: (D)

PART B · SELF-ASSESSMENT — 23 PROBLEMS

36. MEDIUM · DESMOS

The function h is defined by $h(x) = 2x^2 - 12x + 7$. What is the x-coordinate of the vertex of the graph of $y = h(x)$ in the xy-plane?

(A) 3 ✓ (B) 6

(C) -3 (D) 2

1. Type `y=2x^2-12x+7`.

2. Tap the vertex: (3, −11).

3. Check with the axis formula: type `12/(2*2)`; it returns 3. The x-coordinate is 3.

Answer: (A)

37. MEDIUM · DESMOS

The function f is defined by $f(x) = a \cdot 3^x$, where a is a constant. If $f(2) = 45$, what is the value of $f(4)$?

(A) 225 **(B) 405 ✓**

(C) 450 (D) 135

1. Type `y=a*3^x` (slider a) and plot the point `(2,45)`.

2. Slide a until the curve passes through the point: $a = 5$.

3. Type `5*3^4`: it returns 405.

Answer: (B)

38. MEDIUM · DESMOS

If $f(x) = x^2 - 4x + k$ and $f(1) = 9$, what is the value of k?

(A) 6 (B) 9

(C) 12 ✓ (D) 14

1. Put $x = 1$ into the rule: $1 - 4 + k = 9$. Let x stand for k: type `y=1-4+x` and `y=9`.

2. Tap the intersection: (12, 9).

3. So $k = 12$. Check: type `1^2-4*1+12`; it returns 9.

Answer: (C)

39. Medium · Desmos

Let $f(x) = ax^2$ and $g(x) = bx^4$ for any value of x, where a and b are positive constants. For how many values of x is $f(x) = g(x)$?

(A) Three ✓ (B) None

(C) One (D) Two

1. Pick any positive constants, for example $a = 2$ and $b = 3$: type `y=2x^2` and `y=3x^4`.

2. Tap every intersection: $(0, 0)$ and two more, one on each side of the y-axis.

3. The graphs meet three times, so $f(x) = g(x)$ for three values of x.

Answer: (A)

40. Medium · Desmos

The variable y varies inversely with x. When $x = 6$, $y = 15$. What is the value of y when $x = 10$?

(A) 25 (B) 5

(C) 6 **(D) 9 ✓**

1. Inverse variation means xy is constant: type `6*15`; it returns 90.

2. Type `y=90/x` and tap the point with $x = 10$: $(10, 9)$.

3. So $y = 9$.

Answer: (D)

41. Medium · Desmos

If $f(x) = x^2 + 1$ and $g(x) = 2x - 3$, what is the value of $f(g(2))$?

(A) 5 (B) 1

(C) 2 ✓ (D) 3

1. Type `g(x)=2x-3` and `f(x)=x^2+1`.

2. Type `f(g(2))`: Desmos works from the inside out and returns 2.

3. So $f(g(2)) = 2$.

Answer: (C)

42. Medium · No calculator needed

The function C gives the Celsius equivalent of a temperature reading x, and the function P gives the predicted crop yield, in tons, at temperature x°C for a greenhouse. Suppose $C(30) = 30$ and $P(30) = 18$. What is the value of $P(C(30))$?

(A) 30

(B) 18 ✓

(C) 48

(D) 540

1. Work from the inside out: $C(30) = 30$.

2. Then $P(C(30)) = P(30) = 18$.

3. Type `18` as the final check: the input 30 goes into P unchanged.

Answer: (B)

43. Medium · Desmos

For which value of x is the function $f(x) = \dfrac{2x-1}{x^2-2x-24}$ undefined?

(A) 2

(B) 5

(C) -6

(D) -4 ✓

1. The function is undefined where the denominator is zero. Type `y=x^2-2x-24`.

2. Tap the zeros: $x = -4$ and $x = 6$.

3. Only -4 appears among the options.

Answer: (D)

44. Medium · Desmos

The function h is defined by $h(x) = 3x^2 - 24x + 10$. What is the x-coordinate of the vertex of the graph of $y = h(x)$ in the xy-plane?

(A) 4 ✓

(B) 8

(C) -4

(D) 3

1. Type `y=3x^2-24x+10`.

2. Tap the vertex: $(4, -86)$.

3. The x-coordinate is 4. Check: type `24/(2*3)`; it returns 4.

Answer: (A)

45. Medium · Desmos

During a scheduled maintenance check, the water level change in a water reservoir is modeled by $f(x) = x^3 - 5x + 12$, where $f(x)$ is the change in level, in centimeters, and x is the number of hours after maintenance begins, for $-3 \le x \le 3$. What is the value of $f(-2)$?

(A) 10

(B) 14 ✓

(C) 18

(D) 6

1. Type `f(x)=x^3-5x+12`.

2. Type `f(-2)`: it returns 14.

3. So $f(-2) = 14$.

Answer: (B)

46. Medium · Desmos

A decorative fountain shoots a stream of water whose height, in meters, is modeled by $h(t) = -5t^2 + 20t$, where t is the time in seconds after the water leaves the nozzle. What is the maximum height, in meters, reached by the stream of water?

(A) 10

(B) 15

(C) 20 ✓

(D) 25

1. Type `y=-5x^2+20x`.

2. Tap the vertex: (2, 20).

3. The maximum height is 20 meters.

Answer: (C)

47. Medium · Desmos

If $f(x) = 2x^2 - 5x + 3$, what is the value of $f(4)$?

(A) 15 ✓

(B) 17

(C) 11

(D) 13

1. Type `f(x)=2x^2-5x+3`.

2. Type `f(4)`: it returns 15.

3. So $f(4) = 15$.

Answer: (A)

48. Medium · Desmos

A farmer has 80 meters of fencing to enclose a rectangular plot, with one side against an existing wall (so fencing is needed only on the other three sides). If x represents the width of the plot (perpendicular to the wall), which function $A(x)$ models the area of the plot, in square meters?

(A) $A(x) = 80x - x^2$

(B) $A(x) = 40x - 2x^2$

(C) $A(x) = 80 - 2x^2$

(D) $A(x) = 80x - 2x^2$ ✓

1. Two widths plus the length use 80 meters, so the length is $80 - 2x$. Type `y=x(80-2x)`.

2. Type each option and hide the others in turn: `y=80x-2x^2` is drawn exactly on top of it.

3. Only that option gives the same graph as the area $x(80 - 2x)$.

Answer: (D)

49. Medium · Desmos

The function f is defined by $f(x) = (-5)(3)^x + 7$. What is the y-intercept of the graph of $y = f(x)$ in the xy-plane?

(A) $(0, -12)$

(B) $(0, -5)$

(C) $(0, 2)$ ✓

(D) $(0, 7)$

1. Type `y=-5*3^x+7`.

2. Tap the point where the curve crosses the y-axis: (0, 2).

3. The y-intercept is $(0, 2)$.

Answer: (C)

50. Medium · Desmos

x and y^2 are directly proportional. If $x = 6$ when $y^2 = 8$, what is the value of y^2 when $x = 15$?

1. Direct proportion means $x = ky^2$. Type `6/8`: it returns $k = 0.75$.

2. Type `15/0.75`: it returns 20.

3. So $y^2 = 20$ when $x = 15$.

Answer: 20

51. Medium · Desmos

Which of the following parabolas has its turning point in the second quadrant of the coordinate plane?

(A) $y = -(x+1)^2 + 2$

(B) $y = (x+2)^2 - 3$ ✓

(C) $y = (x-3)^2 - 1$

(D) $y = -(x+3)^2 - 1$

1. Type all four parabolas, one per line.
2. Tap each vertex: $(-2, -3)$, $(3, -1)$, $(-3, -1)$, $(-1, 2)$.
3. The second quadrant has $x < 0$ and $y > 0$: only $(-1, 2)$ fits.
Answer: (B)

52. Medium · Desmos

The graph of $y = 2x^2 + 12x + 11$ in the xy-plane is a parabola. What are the coordinates of its vertex?

(A) $(3, 65)$

(B) $(-3, 11)$

(C) $(-6, 11)$

(D) $(-3, -7)$ ✓

1. Type `y=2x^2+12x+11`.
2. Tap the vertex: $(-3, -7)$.
3. The vertex is $(-3, -7)$.
Answer: (D)

53. Medium · Desmos

The functions f and g are defined by $f(x) = 2x - 3$ and $g(x) = x^2 + 1$. Which expression is equivalent to $g(f(x))$?

(A) $4x^2 - 12x + 10$ ✓

(B) $4x^2 - 12x + 9$

(C) $2x^2 - 1$

(D) $4x^2 + 10$

1. Type `f(x)=2x-3` and `g(x)=x^2+1`, then `y=g(f(x))`.
2. Type each option as its own line: `y=4x^2-12x+10` lies exactly on top of the composite.
3. Only that expression gives the same graph.
Answer: (A)

54. Medium · Desmos

The function g is defined by $g(x) = \dfrac{5}{(x-2)^2 - 9}$. For which values of x is $g(x)$ undefined?

(A) $x = 11$ only

(B) $x = 2$ only ✓

(C) $x = -1$ and $x = 5$

(D) $x = -5$ and $x = 1$

1. It is undefined where $(x-2)^2 - 9 = 0$. Type `y=(x-2)^2-9`.
2. Tap the zeros: $x = -1$ and $x = 5$.
3. So g is undefined at $x = -1$ and $x = 5$.
Answer: (B)

55. Medium · Desmos

The function $h(x) = a^x + b$, where $a > 0$ and b is a constant, passes through $(0, 10)$ and $(2, 13)$. What is the value of ab?

(A) 20

(B) 26

(C) 13 ✓

(D) 18

1. From $(0, 10)$: $1 + b = 10$, so $b = 9$. Type `y=a^x+9` (slider a) and plot `(2,13)`.
2. Slide a until the curve passes through the point: $a = 2$.
3. Type `2*9`: it returns $ab = 18$.
Answer: (C)

56. Medium · Desmos

The function $h(x) = 2(x-4)^2 - 32$ has x-intercepts at $(0, 0)$ and $(t, 0)$. What is the value of t?

1. Type `y=2(x-4)^2-32`.
2. Tap the two x-intercepts: (0, 0) and (8, 0).
3. So $t = 8$.
Answer: 8

57. Medium · Desmos

$g(x) = \dfrac{2x - 4}{(x+11)(x-6)}$ If $g(a+5) = 0$, what is the value of a?

1. Type `y=(2x-4)/((x+11)(x-6))`.
2. Tap the x-intercept: (2, 0), so $g(2) = 0$ and $a + 5 = 2$.
3. Type `2-5`: it returns $a = -3$.
Answer: -3

58. HARD · DESMOS

A town's population is modeled by $P(t) = 50(3)^{\frac{t}{4}}$, where t is the number of years since 2010. According to the model, in how many years after 2010 will the population be 1,350?

1. Type `y=50*3^(x/4)` and the line `y=1350`.

2. Tap the intersection: (12, 1350).

3. The population reaches 1,350 after 12 years.

Answer: 12